Machine-Learning-Based Near-Urban Expansion Suitability Analysis

Enugu State, Nigeria - Leakage-Free Reconstruction and Spatial Validation

Abstract

This study reconstructs a machine-learning-based near-urban expansion suitability analysis for Enugu State, Nigeria, after a methodological audit identified target leakage and a distance-driven control design in the earlier workflow. The corrected experiment asks whether seven baseline/static environmental and accessibility predictors can distinguish observed 2020–2025 expansion from stable non-built controls within the same 30–90 m near-urban support. Extra Trees was evaluated using spatially separated Train, Validation and Test blocks and benchmarked against a trivial distance-to-existing-built-up model. On the independent spatial Test set, Extra Trees achieved ROC-AUC 0.727, PR-AUC 0.332, balanced accuracy 0.661 and F1 0.337. The distance-only benchmark achieved ROC-AUC 0.707 and PR-AUC 0.190; thus Extra Trees improved discrimination by +0.020 ROC-AUC and +0.142 PR-AUC, although the baseline retained slightly higher balanced accuracy. The final probability surface uses a Validation-selected operating threshold of 0.36. High-confidence predictions at p>=0.70 cover only 37.27 km² (4.25% of the valid prediction domain), indicating that the model assigns strong confidence to a limited portion of the near-urban landscape. This sparse high-confidence result is retained explicitly rather than hidden through threshold adjustment.

1. Project Background and Research Question

Urban expansion suitability models can appear highly accurate when spatially autocorrelated samples, target-derived predictors or sampling rules inadvertently encode the outcome. The Enugu project was therefore reconstructed from the label-design stage onward. The final research question is: among comparable near-urban pixels within the observed 30-90 m proximity support to 2020 built-up land, can baseline environmental, accessibility, population and land-cover characteristics distinguish pixels that transitioned to built-up by 2025?

2. Leakage Audit and Target Reconstruction

The audit identified two major problems in the earlier experiment: administrative/sample-construction fields could enter the predictor set, and the original positive and negative samples occupied nearly disjoint distance ranges relative to 2020 built-up land. The corrected design reconstructs stable non-built controls from the same 30-90 m support. Distance to 2020 built-up is excluded from the ML predictors and used only as a benchmark.

3. Data and Predictors

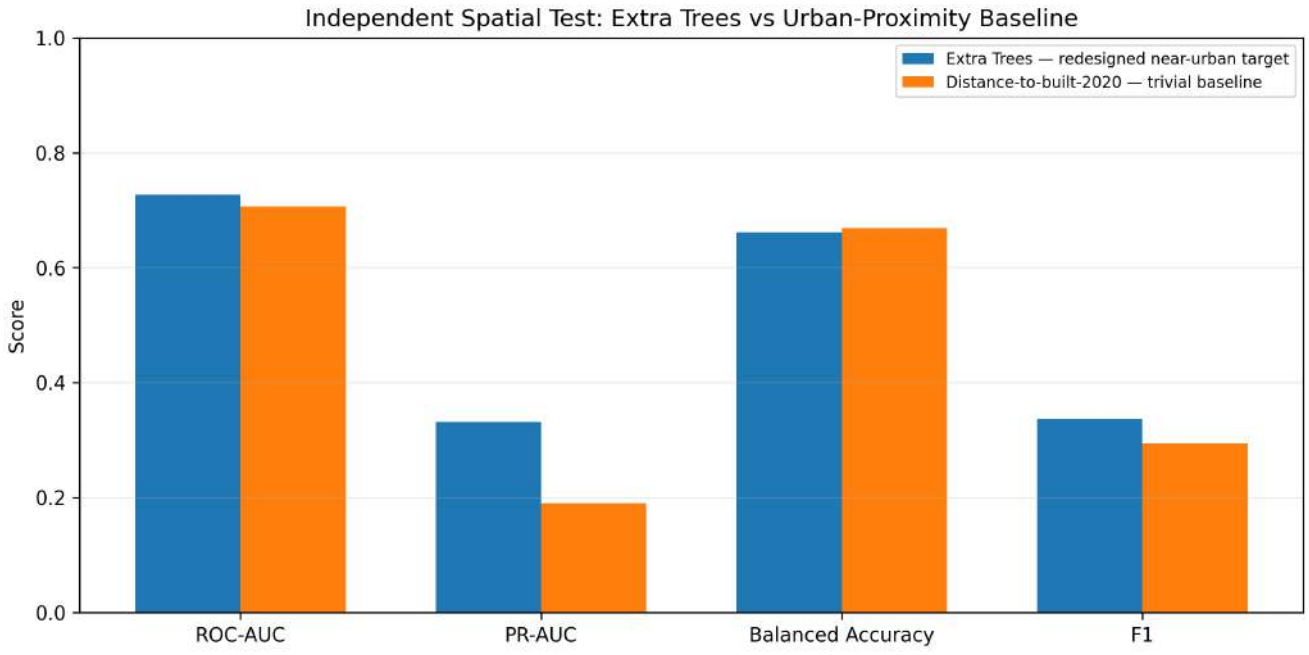
The final model uses elevation, slope, distance to road, distance to recurring surface water, distance to drainage, 2020 population density and baseline 2020 land cover. Outcome-year variables and administrative/spatial-index fields are excluded.

Predictor	ExtraTrees_Importance
Population_Density	0.2862
Distance_to_Surface_Water_m	0.1919
Elevation_m	0.1836
Distance_to_Road_m	0.1202
Baseline_LULC_2020	0.1117
Distance_to_Drainage_m	0.0739
Slope_deg	0.0325

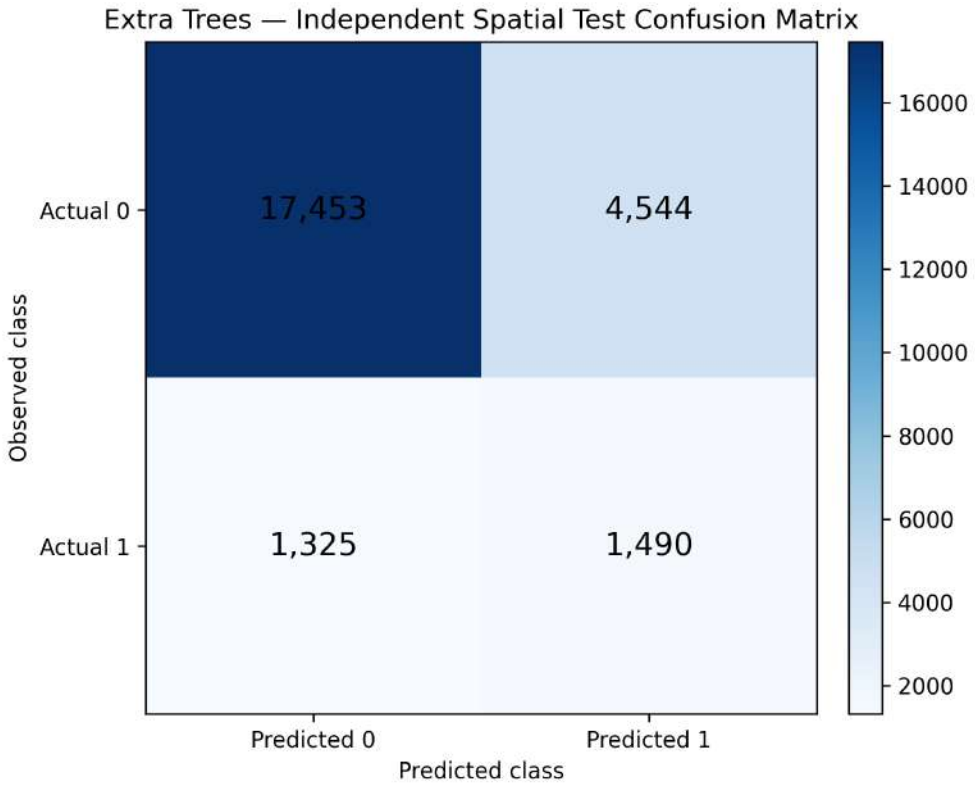
4. Spatial Validation and Baseline Benchmark

Spatial blocks were used to separate Train, Validation and Test partitions. The Test blocks were retained for final independent evaluation. The distance-to-built baseline was evaluated under the same redesigned near-urban prediction question.

Model	ROC_AUC	PR_AUC	Balanced_Accuracy	Precision	Recall	F1
Extra Trees — redesigned near-urban target	0.7267	0.3319	0.6614	0.2469	0.5293	0.3368
Distance-to-built-2020 — trivial baseline	0.7068	0.1903	0.6689	0.1790	0.8178	0.2937



Extra Trees improves ROC-AUC and PR-AUC, while the trivial distance baseline retains slightly higher balanced accuracy. The final interpretation therefore emphasizes added discrimination rather than dominance on every metric.

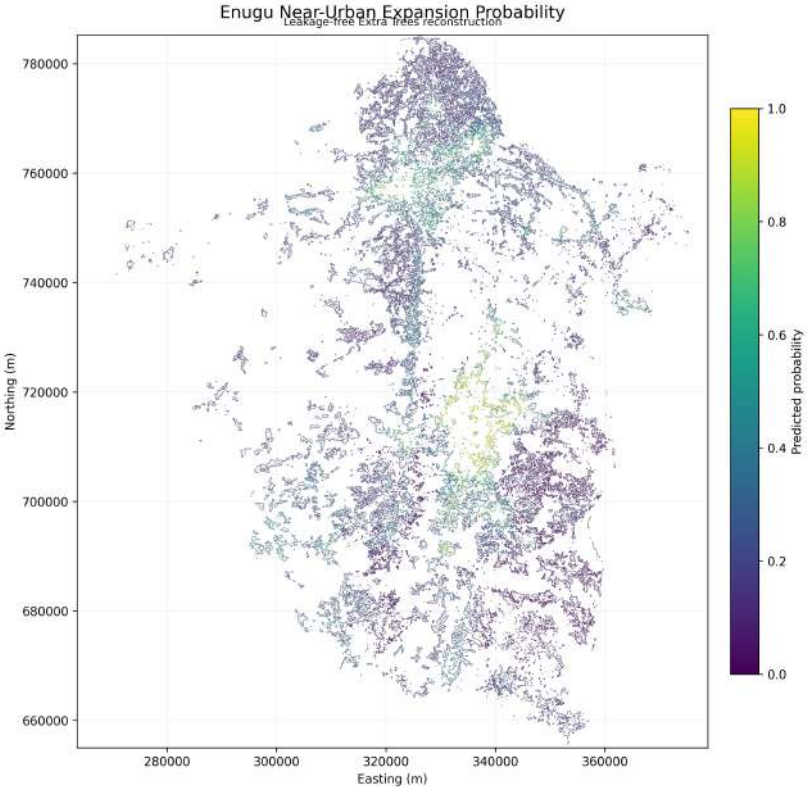


Threshold = 0.36, selected on Validation only. The spatial Test set was used only for final evaluation.

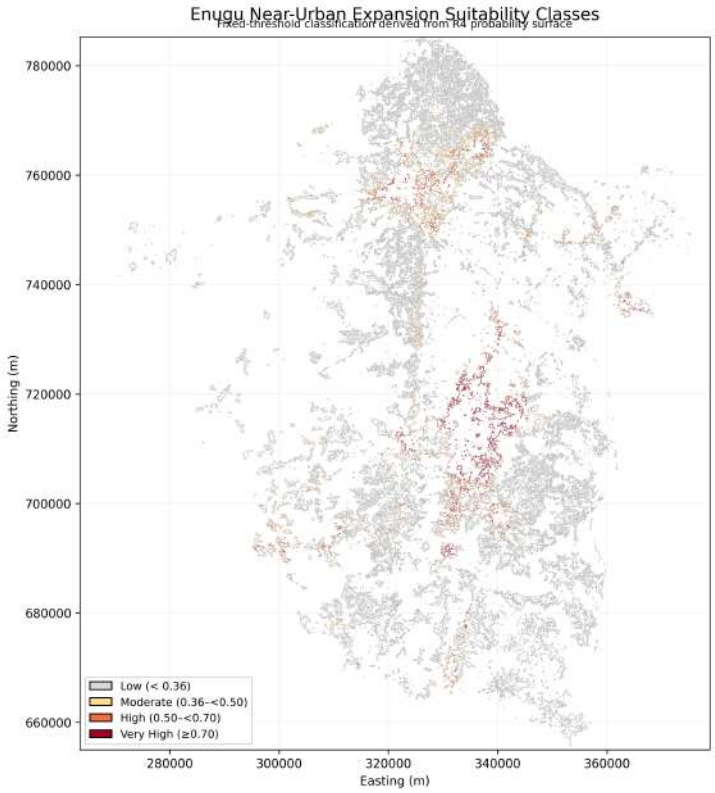
5. Final Probability Surface and Suitability Classes

The Validation-selected operating threshold is 0.36. Final classes use fixed numerical thresholds rather than quantiles.

Suitability_Class	Area_km2	Pct_of_Valid_Prediction_Domain
Low	715.1283	81.5286
Moderate	76.4901	8.7203
High	48.2643	5.5024
Very High	37.2672	4.2487



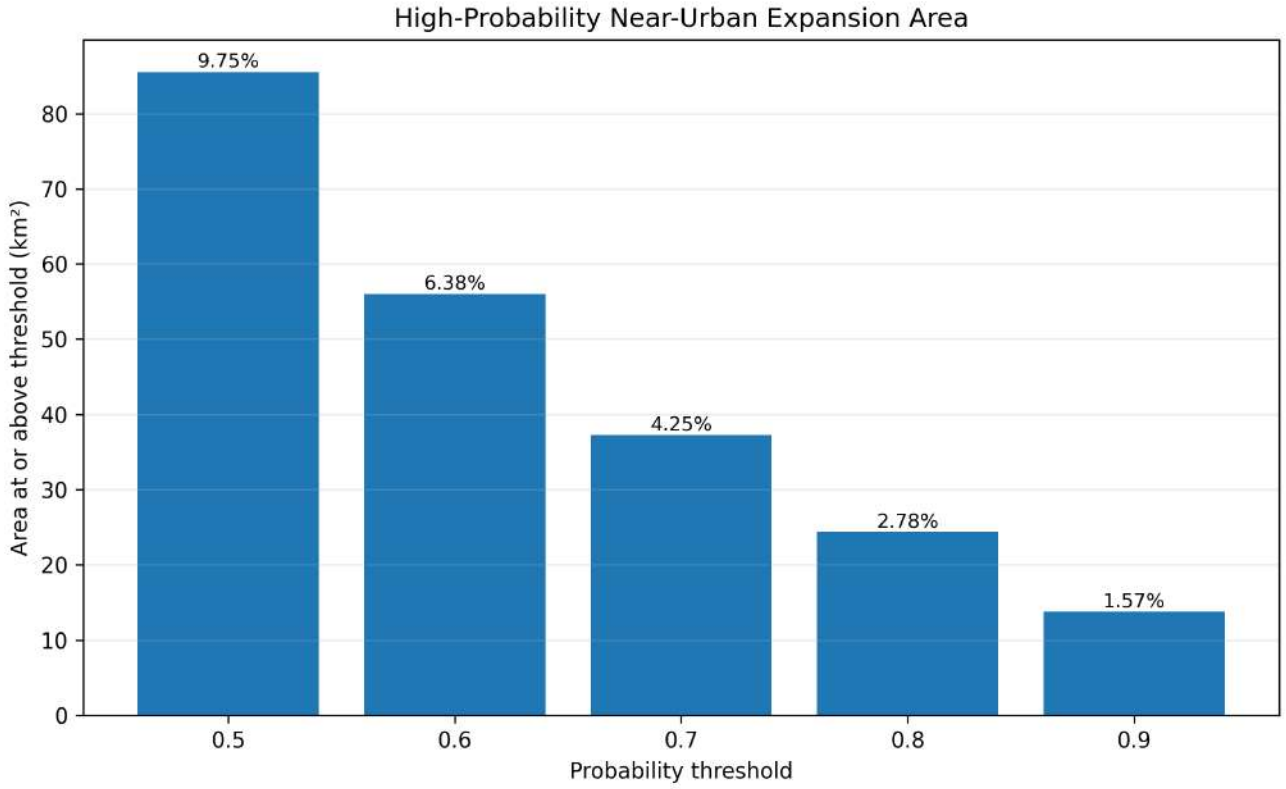
Model: leakage-free Extra Trees using seven baseline/static predictors. Target: observed 2020–2025 expansion among near-urban candidate pixels. Independent spatial Test ROC-AUC = 0.7267; PR-AUC = 0.3319. Validation-selected operating threshold = 0.36. Distance-to-built-2020 was excluded from ML predictors and retained only as a benchmark.



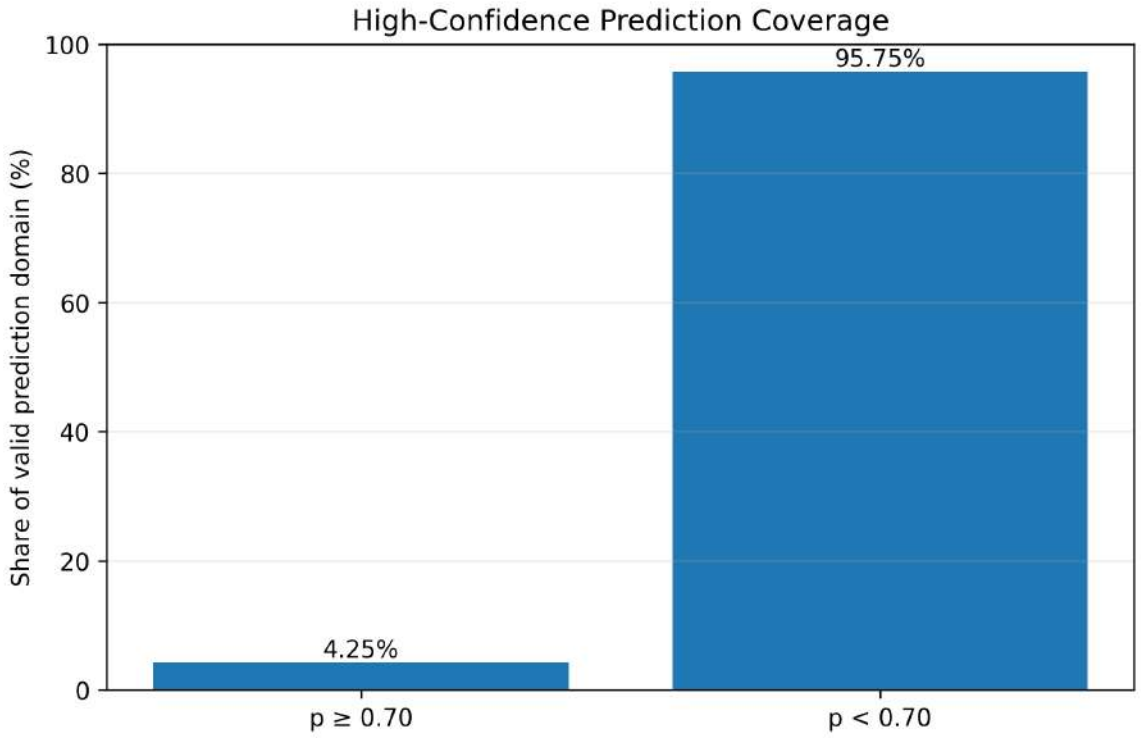
Fixed numerical probability thresholds; no quantile or equal-area classification. Very High (p≥0.70) = 37.27 km² (4.25% of valid prediction domain). Sparse high-confidence coverage is retained as a substantive model result.

6. Confidence and Threshold Diagnostics

Only 37.27 km² (4.25%) reaches p>=0.70. This sparse high-confidence result is retained explicitly.



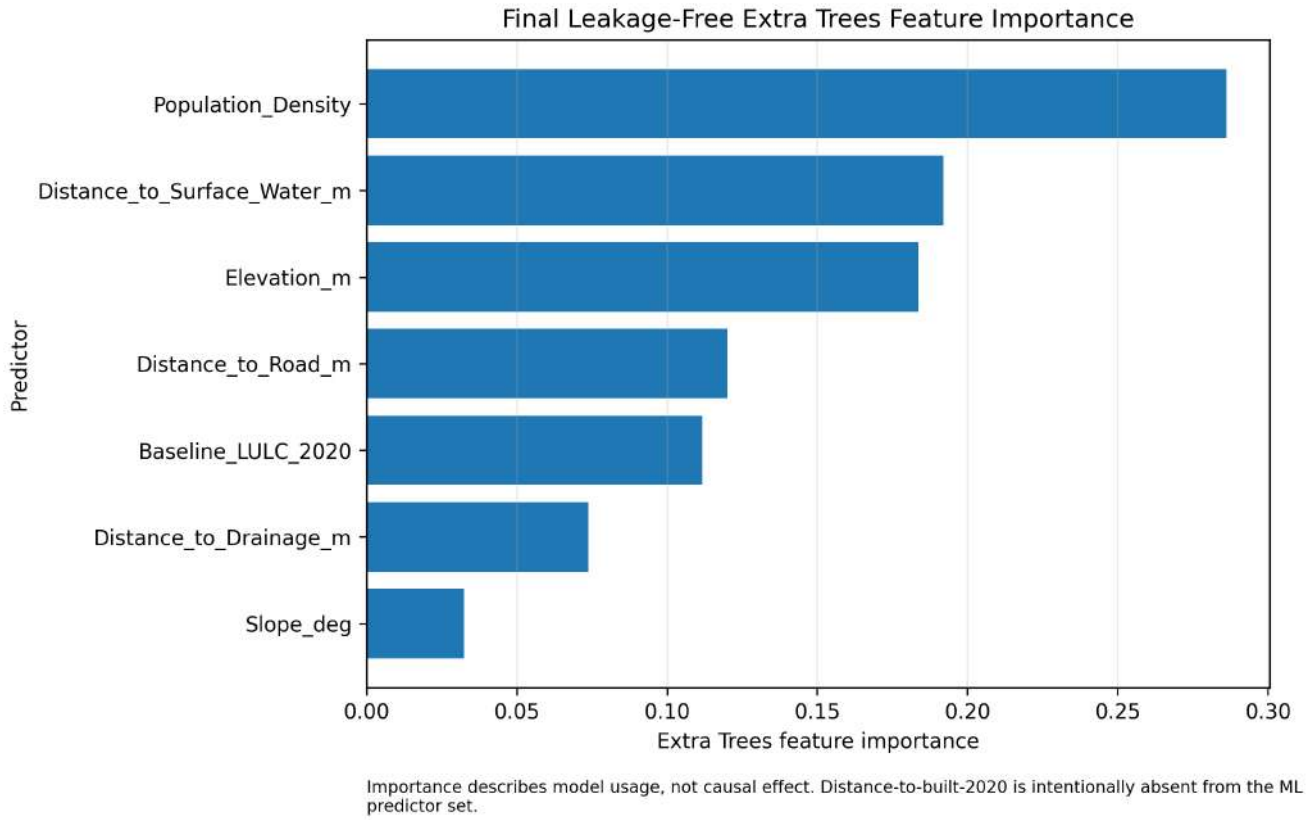
The ≥0.70 threshold is reported as a confidence descriptor, not tuned to enlarge mapped coverage.



Only 37.27 km² (4.25%) reaches p≥0.70. This sparse coverage is retained as a substantive model result rather than hidden by threshold adjustment.

7. Predictor Importance

Feature importance reflects model usage and predictive association, not causal influence.



8. Interpretation

Extra Trees achieved ROC-AUC 0.727 and PR-AUC 0.332. Relative to distance-only proximity, this represents gains of +0.020 ROC-AUC and +0.142 PR-AUC. Because the baseline retained slightly higher balanced accuracy, the final claim is added discrimination rather than superiority on every metric.

9. Limitations

1. The model predicts observed near-urban expansion under the reconstructed 2020–2025 experiment; it should not be interpreted as causal probability, planning approval, or a universal land-suitability score.

2. Independent spatial Test performance is moderate rather than near-perfect: ROC-AUC 0.727, PR-AUC 0.332, and balanced accuracy 0.661.

3. Extra Trees improves discrimination relative to the distance-only benchmark, particularly PR-AUC (+0.142), but the distance baseline retains slightly higher threshold-dependent balanced accuracy (0.669 vs 0.661). Therefore the correct claim is added discrimination, not superiority on every metric.

4. The redesigned target is restricted to a 30–90 m near-urban support. Results should not be extrapolated to the entire state or to locations outside the prediction domain without further validation.

5. Feature importance measures model usage, not causality. Population density, water proximity and elevation should therefore be interpreted as predictive associations.

6. High-confidence p>=0.70 predictions remain sparse: 37.27 km² (4.25% of the valid prediction domain). This result should be reported explicitly rather than hidden by lowering the threshold.

7. The experiment is based on available 2020/2025 land-cover and environmental data; uncertainty in source data propagates into the labels and model outputs.

10. Conclusion

The reconstructed workflow replaces the earlier leakage-contaminated interpretation with a transparent spatially validated experiment. The final package preserves both the useful predictive signal and the substantial uncertainty revealed by the independent spatial Test.